EMBEDDED SYSTEMS & IoT DEVICE DESIGN

INTERNSHIP TASK – 1

Smart Lighting System Using Arduino UNO and LDR

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Introduction

Introduction to Embedded Systems

An embedded system is a computer-based system designed to perform a specific function within a larger electronic or mechanical device. It combines hardware and software to receive inputs, process information, and produce an appropriate output. Embedded systems are generally designed to be reliable, efficient, compact, and suitable for real-time operation.
Embedded systems are used in many areas of our daily lives. They can be found in smart home devices, automobiles, healthcare equipment, industrial machines, robots, washing machines, security systems, and many other electronic products.

Microcontroller vs Microprocessor

A microcontroller is a compact integrated circuit that contains a processor, memory, and input/output peripherals on a single chip. It is commonly used for dedicated control applications. A microprocessor mainly provides processing capability and generally requires external memory and peripherals. Microcontrollers are commonly used in embedded and IoT devices, while microprocessors are widely used in computers and other general-purpose systems.

Component Study

Arduino UNO
Arduino UNO is a popular microcontroller development board based on the ATmega328P. It provides digital and analog input/output pins and can be programmed using the Arduino IDE. It is widely used for learning embedded systems, automation, robotics, and prototyping.
ESP32
ESP32 is a microcontroller platform designed for embedded and IoT applications. It provides built-in Wi-Fi and Bluetooth along with multiple input/output interfaces. It is useful for projects that require wireless communication and remote monitoring.
Raspberry Pi Pico
Raspberry Pi Pico is a compact microcontroller board based on the RP2040 microcontroller. It provides GPIO, analog input, PWM, and USB interfaces and can be used for embedded systems, automation, robotics, and educational projects.
Sensors
A temperature sensor measures the temperature of the surrounding environment. A motion sensor detects movement, commonly using changes in infrared radiation. A light sensor such as an LDR detects changes in light intensity and provides a varying electrical signal that can be read by a microcontroller.

Mini Project

Smart Lighting System

The Smart Lighting System is an embedded system that automatically controls an LED according to the surrounding light intensity. The project uses an Arduino UNO and an LDR sensor. When the environment becomes dark, the LDR produces a lower sensor reading and the Arduino turns the LED ON. When sufficient light is available, the sensor reading increases and the Arduino turns the LED OFF.

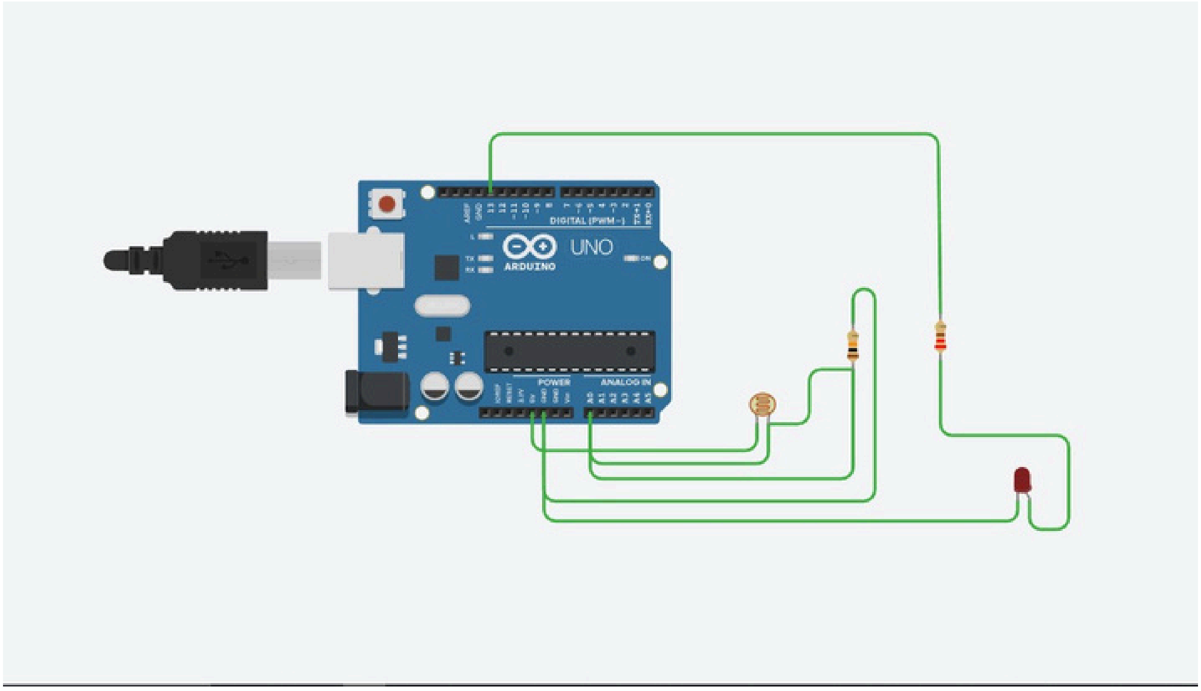
Components Used

- Arduino UNO
- LDR / Photoresistor
- 10 kΩ resistor
- LED
- 220 Ω resistor
- Jumper wires

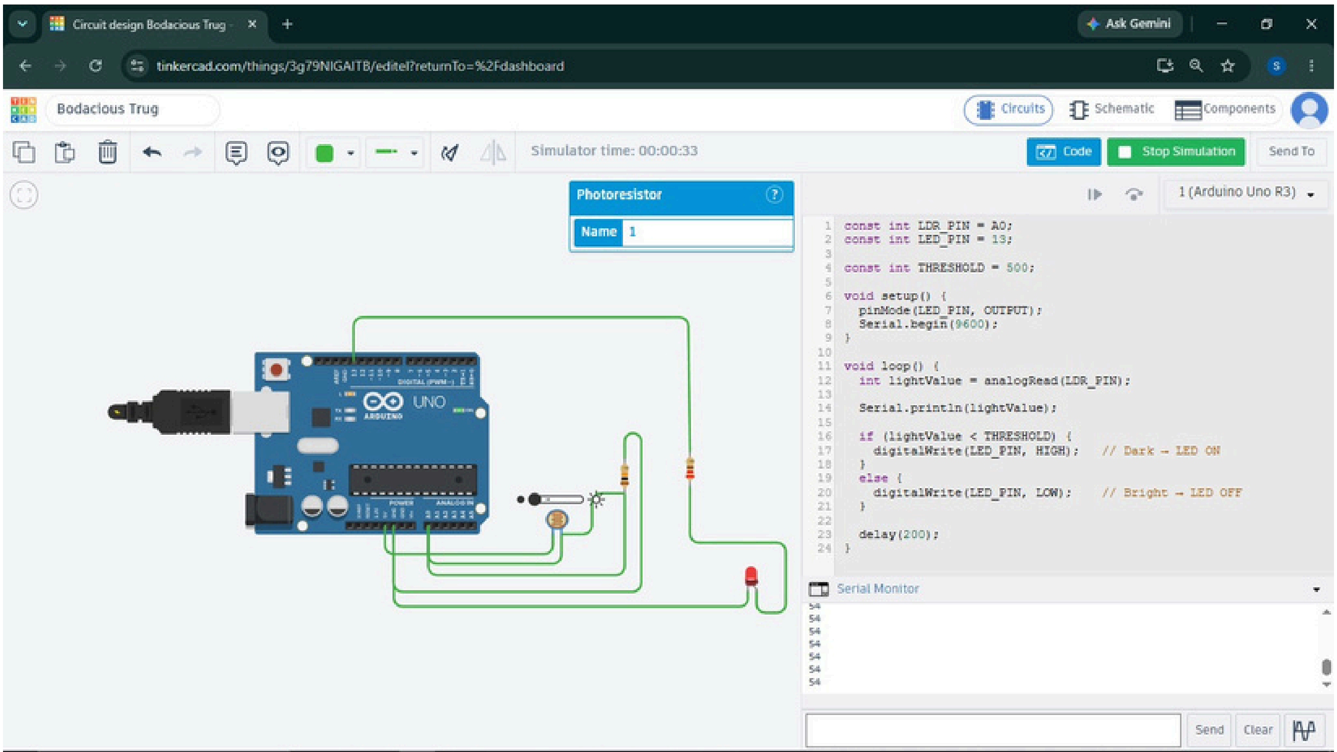
Working Principle

The LDR is connected to the analog input A0 of the Arduino using a voltage divider with a 10 kΩ resistor. The Arduino continuously reads the light intensity from the LDR. The reading is compared with a threshold value of 500. If the reading is below 500, the Arduino considers the environment dark and switches the LED ON. If the reading is 500 or higher, the LED is switched OFF.

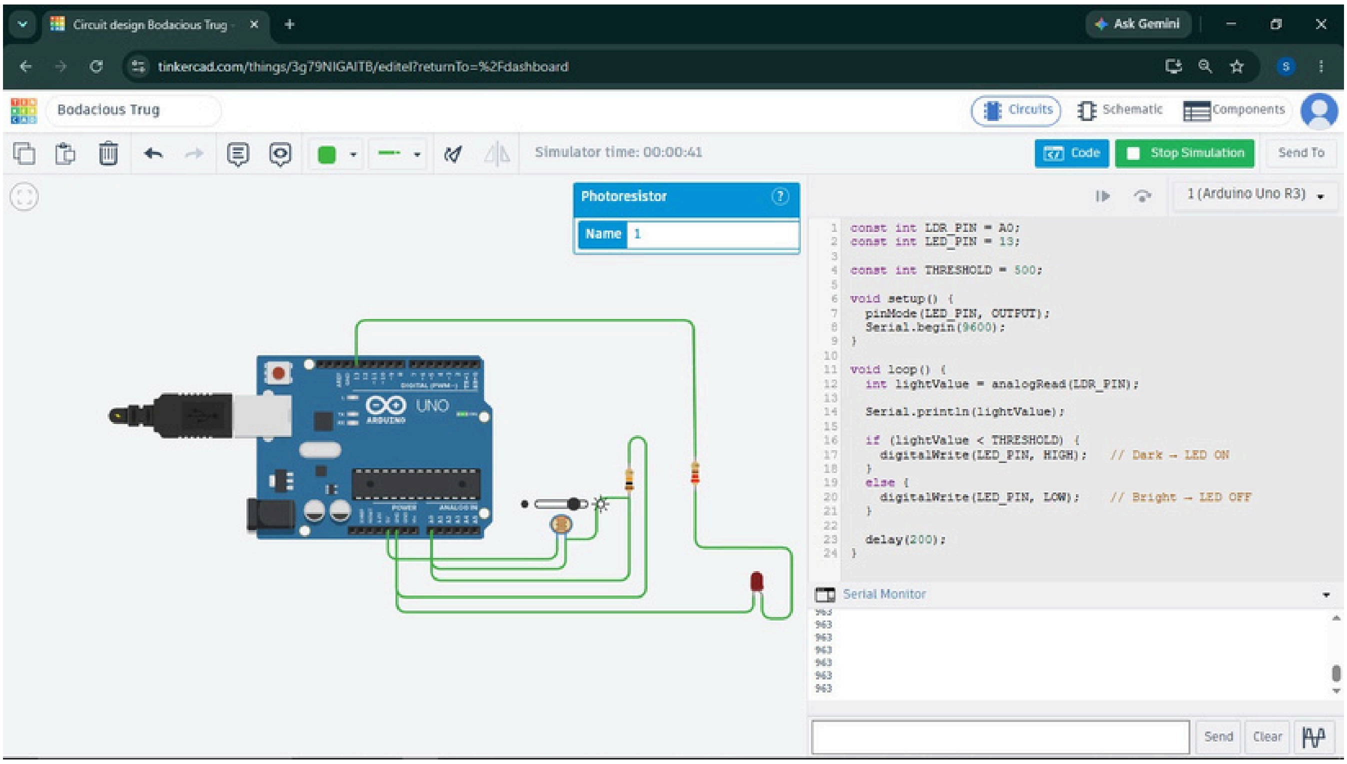
Circuits



UNO
ARDUINO
DIGITAL (PWM ~)
POWER
ANALOG IN



Circuit design Bodacious Trug
Ask Gemini
tinkercad.com/things/3g79NIGAITB/editel?returnTo=%2Fdashboard
Bodacious Trug
Circuits
Schematic
Components
Simulator time: 00:00:33
Code
Stop Simulation
Send To
Photoresistor
Name 1
1 (Arduino Uno R3)
const int LDR_PIN = A0;
const int LED_PIN = 13;
const int THRESHOLD = 500;
void setup() {
pinMode(LED_PIN, OUTPUT);
Serial.begin(9600);
}
void loop() {
int lightValue = analogRead(LDR_PIN);
Serial.println(lightValue);
if (lightValue < THRESHOLD) {
digitalWrite(LED_PIN, HIGH); // Dark → LED ON
}
else {
digitalWrite(LED_PIN, LOW); // Bright → LED OFF
}
delay(200);
}
Serial Monitor
54
54
54
54
54
54
54
Send
Clear



Code

```
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else {
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delay(200);}
```

Results & Applications

Simulation Results

The project was successfully tested using the Tinkercad simulation.
In low-light conditions, the LDR reading was approximately 54, and the LED turned ON. When the light intensity was increased, the reading went above 800, and the LED turned OFF.

Real-World Applications

The Smart Lighting System can be used in automatic street lights, smart homes, gardens, pathways, parking areas, and other locations where lighting needs to be automatically controlled according to the surrounding light conditions.

Conclusion

This project provided practical understanding of embedded systems, microcontrollers, sensors, and basic automation. A Smart Lighting System was successfully designed and simulated using an Arduino UNO and an LDR sensor. The system automatically controls an LED based on the detected light intensity. The successful simulation demonstrates how sensors and microcontrollers can be combined to create simple and useful automated systems.